

Companies want their data to be secure. Rapid increase in the number of sensors and other smart, connected devices has resulted in a parallel explosion of security attacks. This is another factor for the rise of the IIoT, since it provides data security by supporting several latest machine connectivity protocols.

Micro-robotics is another area for the IIoT's improvement. Today, micro-robots are the size of dragon flies, which can be unleashed in huge numbers for gathering data. Newer processors, sensors, integrated wireless connectivity and other features can take the IIoT to greater heights.

Performance of sensors has improved so much over time that these tiny objects can measure several parameters (such as temperature, humidity, velocity, stress, torque, pressure, etc). These are now being integrated with the Internet in many interesting ways. With increased integration, it is now possible to incorporate and embed sensors inside chips. Such chips can support a number of peripherals such as machines, operators and the Internet.

Soon there may be raw materials and supplies that come with embedded tags, which certify the content and quality. This could reduce the time of inspection of incoming materials for the industry.

Today, a bank employee puts his thumb on a fingerprint reader to enter an Internet-based worksheet. In several colleges in India, advanced thumb readers are used for the purpose of attendance analysis.

Imagine a computer numerical control (CNC)-based components manufacturing industry. The same thumb reader may be attached to each CNC machine so that its access is limited to trained and authorised personnel only.

Inspection instruments like digital calipers and micrometers may soon come with tags so that data is automatically transferred to the cloud using analytics software. This could help achieve efficient quality control in the industry.

Today's CNC machines and robots (using common communication protocols and standards) are not

only capable of exchanging data among themselves but are intelligent enough (through the use of sensors and electronic circuits) to determine the health and performance of other machines (such as CNC drilling and milling machines) involved in the manufacturing cycle. These machines have self-correcting mechanisms, and can optimise themselves to avoid wastage and rejection. Thus, the IIoT has reshaped factories. Digitised manufacturing industries adopting the IIoT is referred to as Industry 4.0.

The IIoT is a common term for all hardware that work together through IoT connectivity to enhance manufacturing and other industrial processes. Industrial processes may be core (such as manufacturing and assembly) or non-core (such as inventory control, scheduling, cost analysis and billing).

Protocols such as OPC UA and MQTT are quickly emerging as the standards for the IIoT

However, the IIoT is facing many challenges including security of data. While designing the IIoT for a traditional industrial application, selection of proper protocols is necessary.

The network medium used (high-speed fibre, cable or cellular) is also an important factor in the IIoT. Cellular is gaining popularity in industrial spaces due to its simplicity and secured communication. Industrialists are now looking to create a simple, compact and cost-effective approach for IIoT communication.

Protocols to improve the IIoT

The IIoT follows several protocols for the industry. These can be grouped into two broad categories: client-server and publish-subscribe.

Machines that used edge-of-network technologies are now becoming outdated due to the IIoT. Protocols such as OPC UA and MQTT are quickly emerging as the standards for the IIoT, due to features like lightweight overhead, publish-subscribe model



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